

**Undergraduate Curriculum in Biological Systems Engineering
Food & Bioprocess Engineering Option**

2024-2025 Catalog

Total Credits 128

First Year (33 cr.)

Semester 1 (16 cr.)			Semester 2 (17 cr.)		
R	Engr 1010	Orientation (FS)	1	ABE 1100	Experiencing BSE (S)
3	ABE 1700	Graphics and Design (FS)	3	ABE 1600	Engineering Problems (S)
4	Chem 1670	General Chemistry (FS)	4	Math 1660	Calculus II (FSSS)
1	Chem 1670L	General Chemistry Lab (FS)	4	Phys 2310	Classical Physics I (FSSS)
4	Math 1650	Calculus I (FSSS)	1	Phys 2310L	Classical Physics I Lab (FS)
3	Engl 1500	Crit. Think. and Comm. (FSSS)	3	Engl 2500	Writ/Oral/Vis/Elec Composition (FSSS)
1	Lib 1600	Information Literacy (FSSS)	1	CE 2710	Engineering Foundations of Statics (FSSS)

Second Year (30 cr.)

Semester 3 (15 cr.)			Semester 4 (15 cr.)		
3	ABE 2160	Fund. Ag. and Biol. Engineering (F)	2	ABE 2180	Project Mgmt/Design Ag & Bio Engr (S)
2	CE 2720	Applied Engineering Statics (FS)	1	ABE 2010	Preparing for the Workplace (FS)
3	Biol 2120	Principles of Biology II (FS)	1	ABE 2730	2D Auto CAD (FS)
3	Chem 2310	Elementary Org. Chemistry (FSSS)	3	M E 2310	Thermodynamics (FSSS)
1	Chem 2310L	Elementary Org. Chem Lab (FSSS)	4	Math 2670	Differential Equations (FSSS)
3	→	U. S. Cultures & Communities Elective	3	Micro 3020	Biology of Microorganisms (FSSS)
			1	Micro 3020L	Microbiology Lab (FS)

Third Year (32 cr.)

Semester 5 (16 cr.)			Semester 6 (16 cr.)		
3	ABE 3160	Applications and Syst Modeling (FS)	4	ABE 3630	Electric Power and Electronics (FS)
3	ABE 3780	Mechanics of Fluids (FS)	3	ABE 3800	Principles of BSE (S)
3	FSHN 3110	Food Chemistry (F)	3	ABE 4690	Grain Processing and Handling (S)
1	FSHN 3110L	Food Chemistry Lab (F)	3	EM 3240	Mechanics of Materials (FSSS)
3	STAT 3050	Engineering Statistics (FSSS)	3	I E 3050	Engineering Economic Analysis (FSSS)
1	→	Lab Elective (<i>ABE 3780L preferred</i>)			
3	→	Communication Elective (FSSS)			

Fourth Year (33 cr.)

Semester 7 (17 cr.)			Semester 8 (16 cr.)		
2	ABE 4150	Ag & Biosystems Engr. Design I (FS)	2	ABE 4160	Ag & Biosystems Engr. Design II (FS)
3	ABE 4040	Instrumentation for Ag and Bio Engr (F)	3	ABE 4510	Food and Bioprocess Engineering (S)
3	ABE 4500	Emerging Tech. in Biomanufacturing (F)	4	M E 4360	Heat Transfer (FSSS)
3	ABE 4800	Engineering Analysis of Bio Systems (F)	3	→	SS/H Elective (FSSS)
3	→	Option Elective (F)	3	→	SS/H Elective (FSSS)
3	→	International Perspectives Elect. (FSSS)			

**Please check the current catalog and Schedule of Classes for most recent offerings*